

Creation Date 16-Mar-2018

Revision Date 18-Mar-2024

Revision Number 3

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product identifier

Product Description: **Boron nitride, Aerosol Refractory Paint**
Cat No. : **40389**

Unique Formula Identifier (UFI) **E74Q-A645-HX0R-YFAK**

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use	Laboratory chemicals.
Sector of use	SU3 - Industrial uses: Uses of substances as such or in preparations at industrial sites
Product category	PC21 - Laboratory chemicals
Process categories	PROC15 - Use as a laboratory reagent
Environmental release category	ERC6a - Industrial use resulting in manufacture of another substance (use of intermediates)
Uses advised against	Food, drug, pesticide or biocidal product use

1.3. Details of the supplier of the safety data sheet

Company

Thermo Fisher (Kandel) GmbH
Erlenbachweg 2, 76870 Kandel, Germany
Tel: +49 (0) 721 84007 280
Fax: +49 (0) 721 84007 300

Swiss distributor - Fisher Scientific AG
Neuhofstrasse 11, CH 4153 Reinach
Tel: +41 (0) 56 618 41 11
<https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html>

E-mail address

begel.sdsdesk@thermofisher.com

1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99
CHEMTREC Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:
Tox Info Suisse Emergency Number: **145 (24hr)**
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)
Chemtrec (24h) Toll-Free: 0800 564 402
Chemtrec Local: +41-43 508 20 11 (Zurich)

Poison Centre - Emergency information services

Ireland : National Poisons Information Centre (NPIC) -
01 809 2166 (8am-10pm, 7 days a week)
Malta : +356 2395 2000
Cyprus : +357 2240 5611

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SECTION 2: HAZARDS IDENTIFICATION

2.1. Classification of the substance or mixture

CLP Classification - Regulation (EC) No 1272/2008

Physical hazards

Flammable aerosols

Category 1 (H222)

Health hazards

Serious Eye Damage/Eye Irritation
Specific target organ toxicity - (single exposure)

Category 2 (H319)
Category 3 (H336)

Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

2.2. Label elements



Signal Word

Danger

Hazard Statements

H222 - Extremely flammable aerosol
H229 - Pressurised container: May burst if heated.
H319 - Causes serious eye irritation
H336 - May cause drowsiness or dizziness
EUH066 - Repeated exposure may cause skin dryness or cracking

Precautionary Statements

P211 - Do not spray on an open flame or other ignition source
P251 - Pressurized container: Do not pierce or burn, even after use
P280 - Wear eye protection/ face protection
P312 - Call a POISON CENTER or doctor if you feel unwell
P410 + P412 - Protect from sunlight. Do not expose to temperatures exceeding 50 °C/122 °F

2.3. Other hazards

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB)

This product does not contain any known or suspected endocrine disruptors

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

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3.2. Mixtures

Component	CAS No	EC No	Weight %	CLP Classification - Regulation (EC) No 1272/2008
Acetone	67-64-1	200-662-2	32	Flam. Liq. 2 (H225) Eye Irrit. 2 (H319) STOT SE 3 (H336) EUH066
Ethyl alcohol	64-17-5	200-578-6	20	Flam. Liq. 2 (H225) Eye Irrit. 2 (H319)
Boron nitride (BN)	10043-11-5	EEC No. 233-136-6	20	-
Propane	74-98-6	EEC No. 200-827-9	14	Flam. Gas 1 (H220) Press. Gas (H280)
n-Butane	106-97-8	EEC No. 203-448-7	14	Flam. Gas 1 (H220) Press. Gas (H280)

Component	Specific concentration limits (SCL's)	M-Factor	Component notes
Ethyl alcohol	Eye Irrit. 2 :: C>=50%	-	-

Full text of Hazard Statements: see section 16

SECTION 4: FIRST AID MEASURES

4.1. Description of first aid measures

General Advice	If symptoms persist, call a physician. Do not breathe dust/fume/gas/mist/vapors/spray.
Eye Contact	Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.
Skin Contact	Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.
Ingestion	Clean mouth with water and drink afterwards plenty of water.
Inhalation	Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.
Self-Protection of the First Aider	Remove all sources of ignition. Use personal protective equipment as required.

4.2. Most important symptoms and effects, both acute and delayed

. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: May cause pulmonary edema: Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting: May cause central nervous system depression

4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician	Treat symptomatically. Symptoms may be delayed.
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SECTION 5: FIREFIGHTING MEASURES

5.1. Extinguishing media

Suitable Extinguishing Media

Water spray, carbon dioxide (CO2), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

Extinguishing media which must not be used for safety reasons

Do not use water jetstream.

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5.2. Special hazards arising from the substance or mixture

Flammable. Risk of ignition. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO₂).

5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Fight fire remotely due to the risk of explosion.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

Use personal protective equipment as required. Ensure adequate ventilation. Remove all sources of ignition. Take precautionary measures against static discharges.

6.2. Environmental precautions

Should not be released into the environment.

6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

Pressurized container: Do not pierce or burn, even after use

6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

SECTION 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

Do not get in eyes, on skin, or on clothing. Wear personal protective equipment/face protection. Ensure adequate ventilation. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges. Keep away from heat/sparks/open flames/hot surfaces. - No smoking. Avoid breathing vapors or mists.

Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

7.2. Conditions for safe storage, including any incompatibilities

Flammables area. Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame. Pressurized container: Do not pierce or burn, even after use. Use only outdoors or in a well-ventilated area. Protect from sunlight. Do not expose to temperatures exceeding 50 °C/122 °F.

Technical Rules for Hazardous Substances (TRGS) 510
Storage Class (LGK) (Germany)

Storage Class/LGK 2B

Switzerland - Storage of hazardous substances

Storage class - SC 2
<https://www.kvu.ch/de/themen/stoffe-und-produkte>
<https://www.kvu.ch/fr/themes/substances-et-produits>

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<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

7.3. Specific end use(s)

Use in laboratories

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control parameters

Exposure limits

List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

Component	European Union	The United Kingdom	France	Belgium	Spain
Acetone	TWA: 500 ppm (8h) TWA: 1210 mg/m ³ (8h)	TWA: 500 ppm TWA: 1210 mg/m ³ STEL: 1500 ppm STEL: 3620 mg/m ³	TWA / VME: 500 ppm (8 heures). restrictive limit TWA / VME: 1210 mg/m ³ (8 heures). restrictive limit STEL / VLCT: 1000 ppm. restrictive limit STEL / VLCT: 2420 mg/m ³ . restrictive limit	TWA: 246 ppm 8 uren TWA: 594 mg/m ³ 8 uren STEL: 492 ppm 15 minuten STEL: 1187 mg/m ³ 15 minuten	TWA / VLA-ED: 500 ppm (8 horas) TWA / VLA-ED: 1210 mg/m ³ (8 horas)
Ethyl alcohol		TWA: 1000 ppm TWA; 1920 mg/m ³ TWA WEL - STEL: 3000 ppm STEL; 5760 mg/m ³ STEL	TWA / VME: 1000 ppm (8 heures). TWA / VME: 1900 mg/m ³ (8 heures). STEL / VLCT: 5000 ppm. STEL / VLCT: 9500 mg/m ³ .	TWA: 1000 ppm 8 uren TWA: 1907 mg/m ³ 8 uren	STEL / VLA-EC: 1000 ppm (15 minutos). STEL / VLA-EC: 1910 mg/m ³ (15 minutos).
Propane				TWA: 1000 ppm 8 uren	TWA / VLA-ED: 1000 ppm (8 horas)
n-Butane		STEL: 750 ppm 15 min STEL: 1810 mg/m ³ 15 min TWA: 600 ppm 8 hr TWA: 1450 mg/m ³ 8 hr Carc. containing >0.1% Buta-1,3-diene	TWA / VME: 800 ppm (8 heures). TWA / VME: 1900 mg/m ³ (8 heures).	STEL: 980 ppm 15 minuten STEL: 2370 mg/m ³ 15 minuten	TWA / VLA-ED: 1000 ppm (8 horas)

Component	Italy	Germany	Portugal	The Netherlands	Finland
Acetone	TWA: 500 ppm 8 ore. Time Weighted Average TWA: 1210 mg/m ³ 8 ore. Time Weighted Average	TWA: 500 ppm TWA: 1200 mg/m ³	STEL: 750 ppm 15 minutos TWA: 500 ppm 8 horas TWA: 1210 mg/m ³ 8 horas	STEL: 2420 mg/m ³ 15 minuten TWA: 1210 mg/m ³ 8 uren	TWA: 500 ppm 8 tunteina TWA: 1200 mg/m ³ 8 tunteina STEL: 630 ppm 15 minuutteina STEL: 1500 mg/m ³ 15 minuutteina
Ethyl alcohol		200 ppm TWA MAK; 380 mg/m ³ TWA MAK	STEL: 1000 ppm 15 minutos	huid STEL: 1900 mg/m ³ 15 minuten TWA: 260 mg/m ³ 8 uren	TWA: 1000 ppm 8 tunteina TWA: 1900 mg/m ³ 8 tunteina STEL: 1300 ppm 15 minuutteina STEL: 2500 mg/m ³ 15 minuutteina
Propane		TWA: 1000 ppm (8 Stunden). AGW - exposure factor 4 TWA: 1800 mg/m ³ (8 Stunden). AGW -	TWA: 1000 ppm 8 horas		TWA: 800 ppm 8 tunteina TWA: 1500 mg/m ³ 8 tunteina STEL: 1100 ppm 15

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		exposure factor 4 TWA: 1000 ppm (8 Stunden). MAK TWA: 1800 mg/m ³ (8 Stunden). MAK Höhepunkt: 4000 ppm Höhepunkt: 7200 mg/m ³			minuutteina STEL: 2000 mg/m ³ 15 minuutteina
n-Butane		TWA: 1000 ppm (8 Stunden). AGW - exposure factor 4 TWA: 2400 mg/m ³ (8 Stunden). AGW - exposure factor 4 TWA: 1000 ppm (8 Stunden). MAK TWA: 2400 mg/m ³ (8 Stunden). MAK Höhepunkt: 4000 ppm Höhepunkt: 9600 mg/m ³	STEL: 1000 ppm 15 minutos TWA: 1000 ppm 8 horas		TWA: 800 ppm 8 tunteina TWA: 1900 mg/m ³ 8 tunteina STEL: 1000 ppm 15 minuutteina STEL: 2400 mg/m ³ 15 minuutteina

Component	Austria	Denmark	Switzerland	Poland	Norway
Acetone	MAK-KZGW: 2000 ppm 15 Minuten MAK-KZGW: 4800 mg/m ³ 15 Minuten MAK-TMW: 500 ppm 8 Stunden MAK-TMW: 1200 mg/m ³ 8 Stunden	TWA: 250 ppm 8 timer TWA: 600 mg/m ³ 8 timer STEL: 500 ppm 15 minutter STEL: 1200 mg/m ³ 15 minutter	STEL: 1000 ppm 15 Minuten STEL: 2400 mg/m ³ 15 Minuten TWA: 500 ppm 8 Stunden TWA: 1200 mg/m ³ 8 Stunden	STEL: 1800 mg/m ³ 15 minutach TWA: 600 mg/m ³ 8 godzinach	TWA: 125 ppm 8 timer TWA: 295 mg/m ³ 8 timer STEL: 156.25 ppm 15 minutter. value calculated STEL: 368.75 mg/m ³ 15 minutter. value calculated
Ethyl alcohol	MAK-KZGW: 2000 ppm 15 Minuten MAK-KZGW: 3800 mg/m ³ 15 Minuten MAK-TMW: 1000 ppm 8 Stunden MAK-TMW: 1900 mg/m ³ 8 Stunden	TWA: 1000 ppm 8 timer TWA: 1900 mg/m ³ 8 timer STEL: 2000 ppm 15 minutter STEL: 3800 mg/m ³ 15 minutter	STEL: 1000 ppm 15 Minuten STEL: 1920 mg/m ³ 15 Minuten TWA: 500 ppm 8 Stunden TWA: 960 mg/m ³ 8 Stunden	TWA: 1900 mg/m ³ 8 godzinach	TWA: 500 ppm 8 timer TWA: 950 mg/m ³ 8 timer STEL: 625 ppm 15 minutter. value calculated STEL: 1187.5 mg/m ³ 15 minutter. value calculated
Propane	MAK-KZGW: 2000 ppm 15 Minuten MAK-KZGW: 3600 mg/m ³ 15 Minuten MAK-TMW: 1000 ppm 8 Stunden MAK-TMW: 1800 mg/m ³ 8 Stunden	TWA: 1000 ppm 8 timer TWA: 1800 mg/m ³ 8 timer STEL: 2000 ppm 15 minutter STEL: 3600 mg/m ³ 15 minutter	STEL: 4000 ppm 15 Minuten STEL: 7200 mg/m ³ 15 Minuten TWA: 1000 ppm 8 Stunden TWA: 1800 mg/m ³ 8 Stunden	TWA: 1800 mg/m ³ 8 godzinach	TWA: 500 ppm 8 timer TWA: 900 mg/m ³ 8 timer TWA: 40 ppm 8 timer TWA: 275 mg/m ³ 8 timer STEL: 625 ppm 15 minutter. value calculated STEL: 1125 mg/m ³ 15 minutter. value calculated
n-Butane	MAK-KZGW: 1600 ppm 15 Minuten MAK-KZGW: 3800 mg/m ³ 15 Minuten MAK-TMW: 800 ppm 8 Stunden MAK-TMW: 1900 mg/m ³ 8 Stunden	TWA: 500 ppm 8 timer TWA: 1200 mg/m ³ 8 timer STEL: 1000 ppm 15 minutter STEL: 2400 mg/m ³ 15 minutter	STEL: 3200 ppm 15 Minuten STEL: 7600 mg/m ³ 15 Minuten TWA: 800 ppm 8 Stunden TWA: 1900 mg/m ³ 8 Stunden	STEL: 3000 mg/m ³ 15 minutach TWA: 1900 mg/m ³ 8 godzinach	TWA: 250 ppm 8 timer TWA: 600 mg/m ³ 8 timer TWA: 40 ppm 8 timer TWA: 275 mg/m ³ 8 timer STEL: 312.5 ppm 15 minutter. value calculated STEL: 750 mg/m ³ 15 minutter. value calculated

Component	Bulgaria	Croatia	Ireland	Cyprus	Czech Republic
Acetone	TWA: 600 mg/m ³ STEL : 1400 mg/m ³	TWA-GVI: 500 ppm 8 satima. TWA-GVI: 1210 mg/m ³ 8 satima.	TWA: 500 ppm 8 hr. TWA: 1210 mg/m ³ 8 hr. STEL: 1500 ppm 15 min STEL: 3630 mg/m ³ 15 min	Skin-potential for cutaneous absorption TWA: 500 ppm TWA: 1210 mg/m ³	TWA: 800 mg/m ³ 8 hodinách. Ceiling: 1500 mg/m ³
Ethyl alcohol	TWA: 1000 mg/m ³	TWA-GVI: 1000 ppm 8 satima. TWA-GVI: 1900 mg/m ³ 8 satima.	STEL: 1000 ppm 15 min		TWA: 1000 mg/m ³ 8 hodinách. Ceiling: 3000 mg/m ³
Propane	TWA: 1800.0 mg/m ³		STEL: 3000 ppm 15 min		
n-Butane	TWA: 1900 mg/m ³	TWA-GVI: 600 ppm 8 satima. TWA-GVI: 1450 mg/m ³	TWA: 1000 ppm 8 hr. STEL: 3000 ppm 15 min		

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		8 satima. TWA-GVI: 10 ppm 8 satima. containing >=0.1% Butadiene TWA-GVI: 22 mg/m ³ 8 satima. containing >=0.1% Butadiene STEL-KGVI: 750 ppm 15 minutama. STEL-KGVI: 1810 mg/m ³ 15 minutama.			
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Component	Estonia	Gibraltar	Greece	Hungary	Iceland
Acetone	TWA: 500 ppm 8 tundides. TWA: 1210 mg/m ³ 8 tundides.	TWA: 500 ppm 8 hr TWA: 1210 mg/m ³ 8 hr	STEL: 3560 mg/m ³ TWA: 1780 mg/m ³	TWA: 1210 mg/m ³ 8 órában. AK	TWA: 250 ppm 8 klukkustundum. TWA: 600 mg/m ³ 8 klukkustundum. Ceiling: 500 ppm Ceiling: 1200 mg/m ³
Ethyl alcohol	TWA: 500 ppm 8 tundides. TWA: 1000 mg/m ³ 8 tundides. STEL: 1000 ppm 15 minutites. STEL: 1900 mg/m ³ 15 minutites.		TWA: 1000 ppm TWA: 1900 mg/m ³	STEL: 3800 mg/m ³ 15 percekben. CK TWA: 1900 mg/m ³ 8 órában. AK	TWA: 1000 ppm 8 klukkustundum. TWA: 1900 mg/m ³ 8 klukkustundum. Ceiling: 2000 ppm Ceiling: 3800 mg/m ³
Propane	TWA: 1000 ppm 8 tundides. TWA: 1800 mg/m ³ 8 tundides.		TWA: 1000 ppm TWA: 1800 mg/m ³		TWA: 1000 ppm 8 klukkustundum. TWA: 1800 mg/m ³ 8 klukkustundum. Ceiling: 2000 ppm Ceiling: 3600 mg/m ³
n-Butane	TWA: 800 ppm 8 tundides. TWA: 1500 mg/m ³ 8 tundides.		TWA: 1000 ppm TWA: 2350 mg/m ³	STEL: 9400 mg/m ³ 15 percekben. CK TWA: 2350 mg/m ³ 8 órában. AK	TWA: 500 ppm 8 klukkustundum. TWA: 1200 mg/m ³ 8 klukkustundum. Ceiling: 1000 ppm Ceiling: 2400 mg/m ³

Component	Latvia	Lithuania	Luxembourg	Malta	Romania
Acetone	TWA: 500 ppm TWA: 1210 mg/m ³	TWA: 500 ppm IPRD TWA: 1210 mg/m ³ IPRD STEL: 1000 ppm STEL: 2420 mg/m ³	TWA: 500 ppm 8 Stunden TWA: 1210 mg/m ³ 8 Stunden	TWA: 500 ppm TWA: 1210 mg/m ³	TWA: 500 ppm 8 ore TWA: 1210 mg/m ³ 8 ore
Ethyl alcohol	TWA: 1000 mg/m ³	TWA: 500 ppm IPRD TWA: 1000 mg/m ³ IPRD STEL: 1000 ppm STEL: 1900 mg/m ³			TWA: 1000 ppm 8 ore TWA: 1900 mg/m ³ 8 ore STEL: 5000 ppm 15 minute STEL: 9500 mg/m ³ 15 minute
Boron nitride (BN)	TWA: 6 mg/m ³	TWA: 6 mg/m ³ IPRD hexagonal and cubic			
Propane	TWA: 1000 ppm TWA: 1800 mg/m ³				TWA: 778 ppm 8 ore TWA: 1400 mg/m ³ 8 ore STEL: 1000 ppm 15 minute STEL: 1800 mg/m ³ 15 minute
n-Butane	TWA: 300 mg/m ³				

Component	Russia	Slovak Republic	Slovenia	Sweden	Turkey
Acetone	TWA: 200 mg/m ³ 1763 MAC: 800 mg/m ³	TWA: 500 ppm TWA: 1210 mg/m ³	TWA: 500 ppm 8 urah TWA: 1210 mg/m ³ 8 urah STEL: 2420 mg/m ³ 15 minutah STEL: 1000 ppm 15 minutah	Indicative STEL: 500 ppm 15 minuter Indicative STEL: 1200 mg/m ³ 15 minuter TLV: 250 ppm 8 timmar. NGV TLV: 600 mg/m ³ 8 timmar. NGV	TWA: 500 ppm 8 saat TWA: 1210 mg/m ³ 8 saat
Ethyl alcohol	TWA: 1000 mg/m ³ 2391	Ceiling: 1920 mg/m ³	TWA: 960 mg/m ³ 8 urah	Indicative STEL: 1000	

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	MAC: 2000 mg/m ³	TWA: 500 ppm TWA: 960 mg/m ³	TWA: 500 ppm 8 urah STEL: 1000 ppm 15 minutah STEL: 1920 mg/m ³ 15 minutah	ppm 15 minuter Indicative STEL: 1900 mg/m ³ 15 minuter TLV: 500 ppm 8 timmar. NGV TLV: 1000 mg/m ³ 8 timmar. NGV	
Boron nitride (BN)	TWA: 6 mg/m ³ 0364 TWA: 6 mg/m ³ 0365				
Propane			TWA: 1000 ppm 8 urah TWA: 1800 mg/m ³ 8 urah STEL: 4000 ppm 15 minutah STEL: 7200 mg/m ³ 15 minutah		
n-Butane	TWA: 300 mg/m ³ 0404 MAC: 900 mg/m ³	TWA: 1000 ppm 8 hodinách containing 0.1% or greater Butadiene TWA: 2400 mg/m ³ 8 hodinách containing 0.1% or greater Butadiene STEL: 5000 ppm 15 minútach containing 0.1% or greater Butadiene STEL: 12000 mg/m ³ 15 minútach containing 0.1% or greater Butadiene	TWA: 1000 ppm 8 urah containing >=0.1% Butadiene TWA: 2400 mg/m ³ 8 urah containing >=0.1% Butadiene STEL: 4000 ppm 15 minutah containing >=0.1% Butadiene STEL: 9600 mg/m ³ 15 minutah containing >=0.1% Butadiene		

Biological limit values

List source(s):

Component	European Union	United Kingdom	France	Spain	Germany
Acetone			Acetone: 100 mg/L urine end of shift	Acetone: 50 mg/L urine end of shift	Acetone: 80 mg/L urine (end of shift)
Component	Italy	Finland	Denmark	Bulgaria	Romania
Acetone				Acetone: 80 mg/L urine at the end of exposure or end of work shift	Acetone: 50 mg/L urine end of shift
Component	Gibraltar	Latvia	Slovak Republic	Luxembourg	Turkey
Acetone			Acetone: 80 mg/L urine end of exposure or work shift		

Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS70 General methods for sampling airborne gases and vapours

MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography

MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

Component	Acute effects local (Oral)	Acute effects systemic (Oral)	Chronic effects local (Oral)	Chronic effects systemic (Oral)
Ethyl alcohol 64-17-5 (20)		DNEL = 87 mg/kg bw/d		

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Component	Acute effects local (Dermal)	Acute effects systemic (Dermal)	Chronic effects local (Dermal)	Chronic effects systemic (Dermal)
Acetone 67-64-1 (32)				DNEL = 186mg/kg bw/day
Ethyl alcohol 64-17-5 (20)				DNEL = 343mg/kg bw/day

Component	Acute effects local (Inhalation)	Acute effects systemic (Inhalation)	Chronic effects local (Inhalation)	Chronic effects systemic (Inhalation)
Acetone 67-64-1 (32)	DNEL = 2420mg/m ³			DNEL = 1210mg/m ³
Ethyl alcohol 64-17-5 (20)	DNEL = 1900mg/m ³			DNEL = 950mg/m ³

Predicted No Effect Concentration (PNEC)

See values below.

Component	Fresh water	Fresh water sediment	Water Intermittent	Microorganisms in sewage treatment	Soil (Agriculture)
Acetone 67-64-1 (32)	PNEC = 10.6mg/L	PNEC = 30.4mg/kg sediment dw	PNEC = 21mg/L	PNEC = 100mg/L	PNEC = 29.5mg/kg soil dw

Component	Marine water	Marine water sediment	Marine water Intermittent	Food chain	Air
Acetone 67-64-1 (32)	PNEC = 1.06mg/L	PNEC = 3.04mg/kg sediment dw			

8.2. Exposure controls

Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting/equipment. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

Personal protective equipment

Eye Protection Goggles (European standard - EN 166)

Hand Protection Protective gloves

Glove material	Breakthrough time	Glove thickness	EU standard	Glove comments
Butyl rubber	> 480 minutes	0.5 mm	EN 374 Level 6	As tested under EN374-3 Determination of Resistance to Permeation by Chemicals
Neoprene gloves	< 30 minutes	0.45 mm		

Skin and body protection Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

Respiratory Protection When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

Large scale/emergency use Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced
Recommended Filter type: low boiling organic solvent Type AX Brown conforming to EN371

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Small scale/Laboratory use Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.
Recommended half mask:- Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141
When RPE is used a face piece Fit Test should be conducted

Environmental exposure controls Do not allow material to contaminate ground water system.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties

Physical State	Aerosol Liquid	
Appearance		
Odor	No information available	
Odor Threshold	No data available	
Melting Point/Range	No data available	
Softening Point	No data available	
Boiling Point/Range	No information available	
Flammability (liquid)	No data available	
Flammability (solid,gas)	Not applicable	Liquid
Explosion Limits	No data available	
Flash Point	No information available	Method - No information available
Autoignition Temperature	No data available	
Decomposition Temperature	No data available	
pH	No information available	
Viscosity	No data available	
Water Solubility	Partially miscible	
Solubility in other solvents	No information available	
Partition Coefficient (n-octanol/water)		
Component	log Pow	
Acetone	-0.24	
Ethyl alcohol	-0.32	
Propane	1.09	
n-Butane	2.31	
Vapor Pressure	No data available	
Density / Specific Gravity	No data available	
Bulk Density	Not applicable	Liquid
Vapor Density	No data available	(Air = 1.0)
Particle characteristics	Not applicable (liquid)	

9.2. Other information

VOC Content(%) 80
Explosive Properties Not explosive Vapors may form explosive mixtures with air
Oxidizing Properties Not oxidising

SECTION 10: STABILITY AND REACTIVITY

10.1. Reactivity None known, based on information available

10.2. Chemical stability Stable under normal conditions.

10.3. Possibility of hazardous reactions

Hazardous Polymerization Hazardous polymerization does not occur.

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Hazardous Reactions None under normal processing.

10.4. Conditions to avoid

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. Protect from sunlight and do not expose to temperatures exceeding 50 °C/122 °F.

10.5. Incompatible materials

Strong oxidizing agents.

10.6. Hazardous decomposition products

Carbon monoxide (CO). Carbon dioxide (CO₂).

SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Product Information

(a) acute toxicity;

Oral No data available
Dermal No data available
Inhalation No data available

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Acetone	5800 mg/kg (Rat)	> 15800 mg/kg (rabbit) > 7400 mg/kg (rat)	76 mg/l, 4 h, (rat)
Ethyl alcohol	LD50 = 10470 mg/kg OECD 401 (Rat) 3450 mg/kg (Mouse)	-	LC50 = 117-125 mg/l (4h) OECD 403 (rat) 20000 ppm/10H (rat)
Boron nitride (BN)	-	LD50 > 2000 mg/kg (Rat)	-
Propane	-	-	LC50 > 20000 ppm (Rat) 4h
n-Butane	-	-	658 mg/L (Rat) 4 h

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

(d) respiratory or skin sensitization;

Respiratory No data available
Skin No data available

Component	Test method	Test species	Study result
Acetone 67-64-1 (32)	Guinea Pig Maximisation Test (GPMT)	guinea pig	non-sensitising
Ethyl alcohol 64-17-5 (20)	Mouse Ear Swelling Test (MEST)	mouse	non-sensitising
	OECD Test Guideline 429 Local Lymph Node Assay	mouse	non-sensitising

(e) germ cell mutagenicity; No data available

Component	Test method	Test species	Study result
Acetone 67-64-1 (32)	OECD Test Guideline 471 AMES test	in vivo	negative
	OECD Test Guideline 476 Mammalian Gene cell mutation	in vitro	negative
Ethyl alcohol	AMES test	in vitro	negative

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64-17-5 (20)	OECD Test Guideline 471 ----- Gene cell mutation OECD Test Guideline 476	Bacteria ----- in vitro Mammalian	----- negative
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(f) carcinogenicity;

No data available

There are no known carcinogenic chemicals in this product

Component	EU	UK	Germany	IARC
n-Butane	Carc Cat. 1A			

(g) reproductive toxicity;

No data available

Component	Test method	Test species / Duration	Study result
Ethyl alcohol 64-17-5 (20)	OECD Test Guideline 416 ----- OECD Test Guideline 414	Oral / mouse 2 Generation ----- Inhalation / Rat	NOAEL = 13.8 g/kg/day ----- NOAEC = 16000 ppm

(h) STOT-single exposure;

Category 3

Results / Target organs

Central nervous system (CNS), Respiratory system.

(i) STOT-repeated exposure;

No data available

Target Organs

No information available.

(j) aspiration hazard;

Based on available data, the classification criteria are not met

Symptoms / effects, both acute and delayed

Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting. May cause pulmonary edema. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting. May cause central nervous system depression.

11.2. Information on other hazards

Endocrine Disrupting Properties

Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

SECTION 12: ECOLOGICAL INFORMATION

12.1. Toxicity

Ecotoxicity effects

Component	Freshwater Fish	Water Flea	Freshwater Algae
Acetone	Oncorhynchus mykiss: LC50 = 5540 mg/l 96h Alburnus alburnus: LC50 = 11000 mg/l 96h Leuciscus idus: LC50 = 11300 mg/L/48h Salmo gairdneri: LC50 = 6100 mg/L/24h	EC50 = 8800 mg/L/48h EC50 = 12700 mg/L/48h EC50 = 12600 mg/L/48h	NOEC = 430 mg/l (algae; 96 h)
Ethyl alcohol	Fathead minnow (Pimephales promelas) LC50 = 14200 mg/l/96h	EC50 = 9268 mg/L/48h EC50 = 10800 mg/L/24h	EC50 (72h) = 275 mg/l (Chlorella vulgaris)
Boron nitride (BN)	LC50: > 100 mg/L, 96h static (Oncorhynchus mykiss)		

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Component	Microtox	M-Factor
Acetone	EC50 = 14500 mg/L/15 min	
Ethyl alcohol	Photobacterium phosphoreum:EC50 = 34634 mg/L/30 min Photobacterium phosphoreum:EC50 = 35470 mg/L/5 min	

12.2. Persistence and degradability

Readily biodegradable

Persistence

Persistence is unlikely, based on information available.

Component	Degradability
Acetone 67-64-1 (32)	91 % (28 d) (OECD 301 B)
Ethyl alcohol 64-17-5 (20)	OECD 301E = 94%

12.3. Bioaccumulative potential

Bioaccumulation is unlikely

Component	log Pow	Bioconcentration factor (BCF)
Acetone	-0.24	0.69 dimensionless
Ethyl alcohol	-0.32	No data available
Propane	1.09	No data available
n-Butane	2.31	No data available

12.4. Mobility in soil

The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Will likely be mobile in the environment due to its volatility. Disperses rapidly in air.

12.5. Results of PBT and vPvB assessment

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB).

12.6. Endocrine disrupting properties

Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

12.7. Other adverse effects

Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected substance
This product does not contain any known or suspected substance

SECTION 13: DISPOSAL CONSIDERATIONS

13.1. Waste treatment methods

Waste from Residues/Unused Products

Waste is classified as hazardous. Pressurized container: Do not pierce or burn, even after use. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

Contaminated Packaging

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

European Waste Catalogue (EWC)

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

Other Information

Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer.

Switzerland - Waste Ordinance

Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600

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<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

SECTION 14: TRANSPORT INFORMATION

IMDG/IMO

14.1. UN number UN1950
14.2. UN proper shipping name AEROSOLS
14.3. Transport hazard class(es) 2.1
14.4. Packing group

ADR

14.1. UN number UN1950
14.2. UN proper shipping name Aerosols
14.3. Transport hazard class(es) 2.1
Subsidiary Hazard Class 5F
14.4. Packing group

IATA

14.1. UN number UN1950
14.2. UN proper shipping name AEROSOLS, FLAMMABLE
14.3. Transport hazard class(es) 2.1
14.4. Packing group

14.5. Environmental hazards No hazards identified
14.6. Special precautions for user No special precautions required.
14.7. Maritime transport in bulk according to IMO instruments Not applicable, packaged goods

SECTION 15: REGULATORY INFORMATION

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

Component	CAS No	EINECS	ELINCS	NLP	IECSC	TCSI	KECL	ENCS	ISHL
Acetone	67-64-1	200-662-2	-	-	X	X	KE-29367	X	X
Ethyl alcohol	64-17-5	200-578-6	-	-	X	X	KE-13217	X	X
Boron nitride (BN)	10043-11-5	233-136-6	-	-	X	X	KE-03535	X	X
Propane	74-98-6	200-827-9	-	-	X	X	KE-29258	X	X
n-Butane	106-97-8	203-448-7	-	-	X	X	KE-03751	X	X

Component	CAS No	TSCA	TSCA Inventory notification - Active-Inactive	DSL	NDSL	AICS	NZIoC	PICCS
Acetone	67-64-1	X	ACTIVE	X	-	X	X	X
Ethyl alcohol	64-17-5	X	ACTIVE	X	-	X	X	X
Boron nitride (BN)	10043-11-5	X	ACTIVE	X	-	X	X	X
Propane	74-98-6	X	ACTIVE	X	-	X	X	X
n-Butane	106-97-8	X	ACTIVE	X	-	X	X	X

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

Authorisation/Restrictions according to EU REACH

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Component	CAS No	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
Acetone	67-64-1	-	Use restricted. See item 75. (see link for restriction details)	-
Ethyl alcohol	64-17-5	-	-	-
Boron nitride (BN)	10043-11-5	-	-	-
Propane	74-98-6	-	-	-
n-Butane	106-97-8	-	Use restricted. See item 28. (see link for restriction details) Use restricted. See item 29. (see link for restriction details) Use restricted. See item 75. (see link for restriction details)	-

REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

Seveso III Directive (2012/18/EC)

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements
Acetone	67-64-1	Not applicable	Not applicable
Ethyl alcohol	64-17-5	Not applicable	Not applicable
Boron nitride (BN)	10043-11-5	Not applicable	Not applicable
Propane	74-98-6	Not applicable	Not applicable
n-Butane	106-97-8	Not applicable	Not applicable

Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

National Regulations

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

WGK Classification

Water endangering class = 1 (self classification)

Component	Germany - Water Classification (AwSV)	Germany - TA-Luft Class
Acetone	WGK1	
Ethyl alcohol	WGK1	
Boron nitride (BN)	nwg	
Propane	nwg	
n-Butane	nwg	

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Component	France - INRS (Tables of occupational diseases)
Acetone	Tableaux des maladies professionnelles (TMP) - RG 84
Ethyl alcohol	Tableaux des maladies professionnelles (TMP) - RG 84

Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

Component	Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81)	Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC)	Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure
Acetone 67-64-1 (32)		Group I	
Ethyl alcohol 64-17-5 (20)		Group I	
Propane 74-98-6 (14)	Prohibited and Restricted Substances	Group I	
n-Butane 106-97-8 (14)	Prohibited and Restricted Substances	Group I	

15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has been conducted by the manufacturer/importer

SECTION 16: OTHER INFORMATION

Full text of H-Statements referred to under sections 2 and 3

H319 - Causes serious eye irritation

H336 - May cause drowsiness or dizziness

EUH066 - Repeated exposure may cause skin dryness or cracking

Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

DNEL - Derived No Effect Level

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

NOEC - No Observed Effect Concentration

PBT - Persistent, Bioaccumulative, Toxic

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer Predicted No Effect Concentration (PNEC)

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

POW - Partition coefficient Octanol:Water

vPvB - very Persistent, very Bioaccumulative

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadviser - LOLI, Merck index, RTECS

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (volatile organic compound)

Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and

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hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

Prepared By	Health, Safety and Environmental Department
Creation Date	16-Mar-2018
Revision Date	18-Mar-2024
Revision Summary	New emergency telephone response service provider.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.
COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No
1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2,
Number 3, Chemo (SR 813.11 - Ordinance on Protection against Dangerous Substances and
Preparations).**

Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

End of Safety Data Sheet